

Digital Data-Capture Ecosystem for Traceable OEE Analysis in a Flexible Manufacturing Cell

Abstract—This paper presents a digital data-capture and integration ecosystem for standardized, traceable Overall Equipment Effectiveness (OEE) analysis in a flexible manufacturing cell. The study was motivated by the limitations of the previous manual workflow, which relied on paper-based records, repeated transcription, spreadsheet-dependent consolidation, and delayed reconciliation across production and failure data. To address this problem, a digital ecosystem based on Microsoft Lists, Power Apps, SharePoint, and Power Query was implemented to enable direct data capture, centralized storage, automated transformation, and integrated analysis. The ecosystem was evaluated through a global inconsistency comparison, a two-proportions test, a paired comparison between original and corrected OEE values, and deterministic sensitivity analyses. The results showed that the global inconsistency rate decreased from 43.43% in the 2025 manual system to 2.25% in the 2026 digital system. In addition, the pipeline improved repeatability, traceability, and efficiency of the OEE calculation process. The main contribution of this work is to demonstrate that reliable OEE analysis in a flexible manufacturing cell depends not only on the KPI formula, but on standardized data capture, clear programmed-time logic, and an integrated digital workflow that improves traceability, repeatability, and analytical reliability.

Keywords—Overall Equipment Effectiveness OEE, data capture standardization, manufacturing data traceability.

I. INTRODUCTION

Overall Equipment Effectiveness (OEE) is increasingly used in Industry 4.0 environments not only as a descriptive manufacturing indicator, but also as an input for forecasting, optimization, and production decision support. Recent work on short-term OEE forecasting demonstrates that the indicator can be obtained using advanced analytical models when reliable operational data are available [1].

Similarly, research on process digitization in small and medium-sized manufacturing enterprises shows that OEE analysis can be integrated with value stream mapping to support process visibility and improvement in smart manufacturing contexts [2]. These developments increase the importance of data reliability, since the usefulness of OEE depends not only on the mathematical structure of the indicator, but also on the quality and traceability of the records used to calculate availability, performance, and quality.

Digital manufacturing research has also emphasized the importance of process-level data for modeling, bottleneck identification, and operational analysis. To obtain this data industrial digitalization may be used to obtain access to large sets of process information, but its use in manufacturing remains challenging because of product complexity, human-centered processes, and data-quality limitations [3].

This is particularly relevant in flexible manufacturing cells, where production, failure, and quality information may be captured across different shifts, products, and operator sources. In these environments, the problem is not only whether data exists, but whether the data can be consistently linked, audited, and transformed into trustworthy performance indicators.

The importance of this issue is reinforced by recent data-quality and data-integrity research. Where data quality is identified as a critical condition for decision-making and emphasize dimensions such as accuracy, completeness, consistency, and validity [4].

In industrial cyber-physical systems, it is further highlighted that data integrity and traceability are central requirements for reliable industrial information systems [5]. Together, these studies support the need to treat OEE calculation as a data-governance problem, not only as a performance-measurement problem.

It is important to distinguish the focus of this work from equipment reliability studies commonly conducted in flexible manufacturing systems. While reliability engineering seeks to characterize equipment failures and maintenance performance, the present research addresses the reliability of the OEE measurement process itself. Specifically, it examines how improvements in data integrity, traceability, and standardization influence the consistency and trustworthiness of the resulting performance indicator.

Since improvements in data integrity depend on the mechanisms used to capture and integrate operational information, recent research has also explored low-code/no-code platforms as accessible technologies for digital manufacturing implementation

To collect large volumes of operational data in industrial processes a related stream of research has explored low-code/no-code platforms as an accessible route for digital manufacturing implementation. Where it is demonstrated that students in a learning factory can develop a digital manufacturing system using low-code/no-code platforms, reducing the software-development barrier and supporting smart-manufacturing learning [6].

However, their work focuses primarily on learning outcomes and digital manufacturing system development. The present study differs by focusing on measurement validity: it evaluates whether a digital data-capture and integration ecosystem can reduce inconsistency, improve traceability, and support more reliable OEE analysis in an educational flexible manufacturing cell.

This paper is based on the hypothesis that a standardized digital data-capture and integration ecosystem can reduce

This work used non-identifiable operational production, failure, and quality records. Funding by FlexLab Research Project.

source-record inconsistency, improve traceability, and support more reliable OEE analysis in a flexible manufacturing cell. This hypothesis emerged from the limitations observed in the previous measurement process, where production, failure, and quality data were captured on paper, manually transcribed into spreadsheets, consolidated through formula-dependent files, and corrected only after delayed reconciliation.

As new production cycles were analyzed, the process became increasingly difficult to sustain because the same manual reconstruction effort had to be repeated, while the origin of errors remained difficult to trace. The study was therefore motivated by the need to transform OEE analysis from a manual, weakly traceable workflow into a standardized digital process capable of preserving the analytical path from each source record to the final indicator.

The contribution of this work is the empirical validation of that transition, showing how data-capture standardization and digital integration can improve data integrity and indicator reliability in an educational flexible manufacturing cell. Unlike previous studies that focus on OEE calculation, forecasting, visualization, or equipment reliability, this work evaluates the reliability of the measurement pipeline itself by validating a digital ecosystem designed to improve data integrity, traceability, and repeatability.

II. METHODS

A. Study context

The study was conducted in an educational flexible manufacturing cell used for production and academic training activities. OEE was monitored through its three standard components: Availability, Performance, and Quality. The calculation model for the study is defined in section II-E.

In the original workflow, production, failures, and quality observations were recorded separately by shift and later consolidated for analysis. Over time, this process evolved from a paper-based and spreadsheet-dependent system into a digital ecosystem based on Microsoft Lists, Power Apps, SharePoint, and Power Query.

B. Problem diagnosis

In 2025 the first version of capturing OEE consisted of these stages:

1. Raw operational data was captured by operators on paper forms.
2. Handwritten data was later transcribed manually into spreadsheet files by team members.
3. Additional calculated columns were added to obtain derived indicators.
4. The resulting document was analyzed manually and necessary corrections to the document were made.

After the first round of analysis, it was discovered that the time and effort to go through the year's data could not be replicated easily due to the high consolidation time and heavy formula management.

C. Iterative Development

The digital ecosystem was developed iteratively in response to successive bottlenecks identified in the original OEE measurement process. The first stage consisted of paper-based records, in which production, failure, and quality data were captured manually by operators and later transcribed into spreadsheets. This approach enabled initial OEE calculation, but it generated duplicated effort, delayed error detection, and weak traceability because the origin of inconsistencies could not be clearly attributed to either the source record or the transcription process.

Microsoft Lists was then introduced to structure the production, failure, and quality datasets in a centralized digital repository. This improved data organization and provided a more consistent storage format; however, if operators continued recording information first on paper and later transferring it to the digital list, the double-entry problem remained. To address this limitation, Power Apps was implemented as the direct capture interface, allowing operators to enter production, failure, and quality information directly into the structured digital records.

Once direct capture was established, the remaining bottleneck was the manual consolidation and transformation of the data for OEE analysis. Power Query was therefore incorporated as the integration layer to retrieve the updated records, clean and transform the datasets, merge production, and failure information by analytical unit, and generate the calculated fields required for the OEE structure. Finally, Power BI was implemented as the visualization layer for monitoring and interpretation. In this way, the ecosystem evolved from isolated manual records into a connected data pipeline in which each tool addressed a specific limitation exposed by the previous stage.

D. Digital Ecosystem Architecture and Data Flow

The digital ecosystem was structured around the interaction between the flexible manufacturing cell, the data-capture applications, the centralized data repositories, and the analytical layer used for Overall Equipment Effectiveness calculation.

In this architecture, the physical cell generated the operational events of interest: production records, failure records, and quality records. Power Apps supported standardized direct data capture, Microsoft Lists and SharePoint supported centralized and traceable storage, Power Query automated data transformation and integration, and Excel served as the calculation environment for the indicator. Finally, Power BI acts as the visualization layer for future monitoring and dashboarding. Therefore, the ecosystem connected the physical manufacturing cell with a digital information flow that preserved the path from each operational record to the final performance indicator.

TABLE I. DIGITAL ECOSYSTEM COMPONENTS AND DATA FLOW

<i>Component</i>	<i>Function in the Ecosystem</i>	<i>Data or Output Generated</i>
Flexible manufacturing cell	Generates production, failure, and quality events	Operational source events

Component	Function in the Ecosystem	Data or Output Generated
Power Apps	Direct digital capture interface for operators	Structured production, failure, and quality forms
Microsoft Lists / SharePoint	Centralized storage of captured records	Traceable digital source tables
Power Query in Excel	Data cleaning, transformation, merging, and calculated fields	Consolidated tables for OEE calculation
Excel	Calculation environment	Availability, performance, quality, and final OEE
Power BI	Visualization layer for monitoring and presenting information	Dashboards and performance views

E. Operational definition of inconsistency

A source record was classified as inconsistent when it presented at least one verifiable problem affecting coherence across the production and failure datasets. These include:

- missing production duration,
- missing product identification in failure logs,
- mismatch between production and failure product,
- failure records without associated production,
- programmed-time values inconsistent with the planned-pallet rule,
- records requiring manual remapping to reconcile sources.

F. Calculation Model for OEE and Validation metrics

Overall Equipment Effectiveness was used as the main performance indicator to evaluate the flexible manufacturing cell. The indicator was calculated as OEE was used as the main performance indicator to evaluate the flexible manufacturing cell. The indicator was calculated as:

$$OEE = A \times P \times Q \quad (1)$$

Where (A) is availability, (P) is performance, and (Q) is quality.

Availability was calculated using the available production time and the consolidated downtime associated with failure records:

$$A = \frac{T_{available} - T_{MTTR}}{T_{available}} \quad (2)$$

Where $T_{available}$ is the available production time calculated by the formula in which the factor 24 x 60 converts date-time differences stored in hours into minutes:

$$T_{available} = (\text{production}_{end_j} - \text{production}_{start_j}) \times 24 \times 60 \quad (3)$$

And T_{MTTR} is equal to the sum of the consolidated failure-duration records, in minutes:

$$T_{MTTR} = \sum_{j=1}^{n_f} T_{MTTR_j} \quad (4)$$

Where n_f is the number of failure records associated with the analytical unit. Each failure duration was calculated as the elapsed time between failure identification and resolution:

$$T_{MTTR_j} = (t_{resolved_j} - t_{identified_j}) \times 24 \times 60 \quad (5)$$

Performance was calculated as the ratio between executed pallets and planned pallets:

$$P = \frac{N_{executed}}{N_{planned}} \quad (6)$$

Where $N_{executed}$ is the number of pallets completed and $N_{planned}$ is the number of pallets expected according to the programmed production time:

$$N_{planned} = f(T_{available}) \quad (7)$$

Planned pallets were obtained from the programmed-time rule established for the cell. This relation is important because recording errors in programmed or available time modifies the number of planned pallets and therefore affects the performance component of OEE.

Quality was calculated from the inspected production quantity and the discarded quantity:

$$Q = \frac{N_{sample} - N_{discarded}}{N_{total}} \quad (8)$$

To evaluate data integrity, the global inconsistency rate was calculated for each workflow: using the following formula:

$$IR_y = \frac{E_y}{N_y} \times 100 \quad (9)$$

Where IR_y is the inconsistency rate for year y . E_y is the number of inconsistent records and N_y is the total audited records.

To evaluate whether data correction affected the OEE indicator, paired differences were calculated between the original and corrected 2025 values:

$$d_i = OEE_{1.0_i} - OEE_{2.0_i} \quad (10)$$

Where d_i is the paired difference for observacion i .

Finally, the sensitivity analysis was expressed using normalized OEE values. For each error scenario, the recalculated OEE was compared with the baseline value:

$$\text{Relative OEE} = \frac{OEE'}{OEE_0} \times 100 \quad (11)$$

Where OEE_0 is the baseline and OEE' is the recalculated value under an error scenario.

G. Analytical procedure

The validation procedure was designed to evaluate both the integrity of the source records and the reliability of the calculated OEE values. Table I summarizes the analyses used in the study and their purpose. The two-proportions test was applied to the inconsistency rates defined in (9). The paired comparison between original and corrected OEE values was based on the difference defined in the paired (10). The deterministic sensitivity analyses used the OEE structure in (1) – (8) and the normalized indicator in (11) to evaluate how recording errors propagate through performance and availability.

TABLE II. VALIDATION METHOD OF DIGITAL OEE ECOSYSTEM

<i>Analysis</i>	<i>Purpose</i>
Global inconsistency rate	Quantify source-record inconsistency before and after digitalization
Two-proportions test	Compare inconsistency rates between 2025 and 2026
Anderson-Darling test	Assess normality of paired OEE differences
Paired t-test	Compare original vs. corrected 2025 OEE values
Sensitivity analysis: programmed time	Measure the effect of programmed-time error on performance and OEE
Sensitivity analysis: MTTR	Measure the effect of MTTR error on availability and OEE
Data-governance comparison	Compare manual and digital workflows beyond the KPI

III. RESULTS

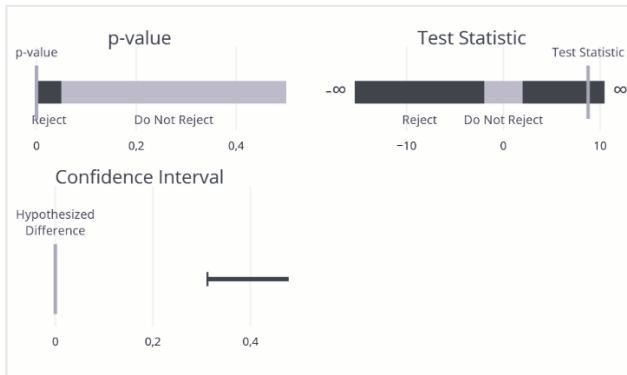
First, the study quantifies the reduction in source-record inconsistency between the manual 2025 workflow and the digital 2026 ecosystem. Second, it evaluates whether the observed data-quality problems affected the OEE values calculated from the 2025 dataset. Finally, deterministic sensitivity analyses are used to explain how recording errors propagate through the OEE structure, specifically through performance in the case of programmed-time error and through availability in the case of MTTR error.

A. Reduction in source-record inconsistency

TABLE III. GLOBAL INCONSISTENCY RATE

Data	System	
	2025 (manual)	2026 (digital)
Production log errors	12	4
Failure log errors	31	0
Total Errors	43	4
Total Records	99	178
Overall Inconsistency Rate	43.43%	2.25%

FIGURE I.2-PROPORTION TEST OEE 2025 VS 2026



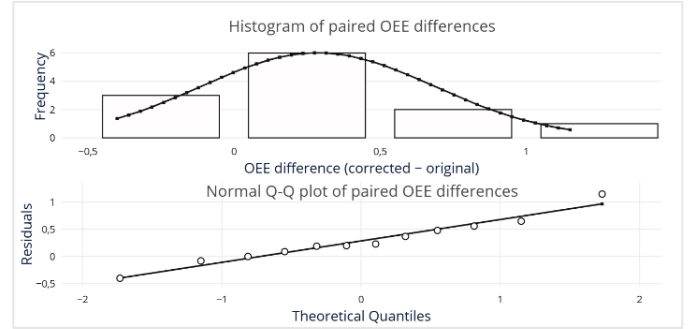
The manual 2025 system exhibited a substantially higher inconsistency rate than the digital 2026 system. This difference was statistically significant according to the two-proportions test, confirming that the digital data-capture ecosystem reduced the proportion of inconsistent source records. Therefore, the proposed workflow not only replaced

manual recording, but also improved the integrity of the analytical input used for OEE calculation.

B. Effect of correction on 2025 OEE values

Before comparing the original and corrected 2025 OEE values, the distribution of the paired differences was evaluated to determine the appropriate paired comparison method.

FIGURE II. NORMALITY ASSESSMENT OF PAIRED OEE DIFFERENCES



The Anderson-Darling result did not reject normality of the paired differences, indicating that the comparison between original and corrected OEE values could be assessed using a parametric paired test.

TABLE IV. DESCRIPTIVE STATISTICS PAIRED T-TEST

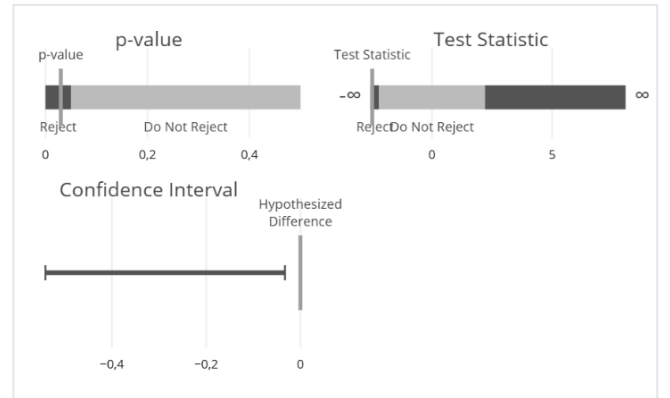
Sample	N	Mean	StDev	SE Mean
OEE 2025 1.0	12	0.4358	0.2728	0.0788
OEE 2025 2.0	12	0.7225	0.3394	0.0980

The paired comparison showed a statistically significant difference between the original and corrected OEE values.

TABLE V. ESTIMATION FOR PAIRED DIFFERENCE

Mean	StDev	SE Mean	95% CI for population mean of (OEE 2025 1.0 - OEE 2025 2.0)	T-Value	P-Value
-0.287	0.400	0.115	(-0.541, -0.033)	-2.48	0.03

FIGURE III. PAIRED T-TEST OEE 2025 1.0 & 2.0

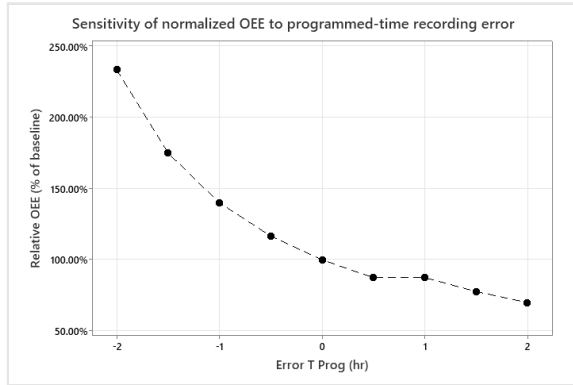


This indicates that the problems present in the manual 2025 workflow were not limited to documentation burden or weak traceability; they directly affected the numerical value of the performance indicator. More importantly, the effect is relevant at the operational decision level because shift- and product-level OEE values can be distorted even when an aggregated indicator appears acceptable. Therefore, preserving traceability by date, shift, and product is necessary to prevent record-level inconsistencies from being concealed by summarized performance views.

C. Sensitivity of OEE to programmed-time recording error

Because programmed time determines the number of planned pallets, any error in this input propagates directly through the performance component of OEE. A deterministic sensitivity analysis was therefore conducted to quantify how deviations in programmed time recording affect relative OEE.

FIGURE IV. SENSITIVITY OF RELATIVE OEE TO PROGRAMMED-TIME RECORDING ERROR

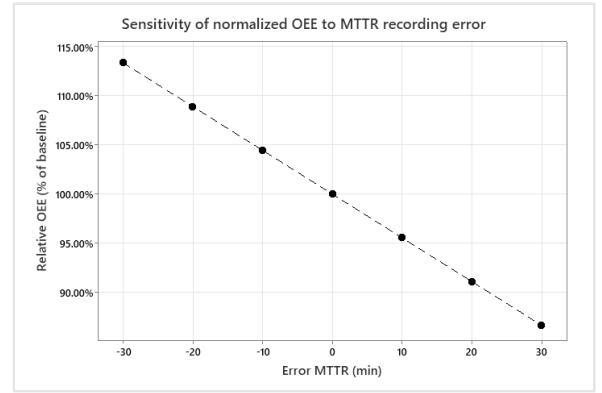


The sensitivity profile shows that underestimation of programmed time artificially inflates OEE by reducing the number of planned pallets and increasing the calculated performance ratio. Conversely, overestimation of programmed time reduces relative OEE. This confirms that programmed-time inconsistency does not primarily distort availability but rather propagates through the performance pathway of the indicator.

D. Sensitivity of OEE to MTTR recording error

A second deterministic sensitivity analysis was used to evaluate how MTTR recording error propagates through the availability component and modifies final OEE.

FIGURE V. SENSITIVITY OF RELATIVE OEE TO MTTR RECORDING ERROR



Unlike programmed-time error, MTTR error affects OEE through availability. The results show a monotonic effect: negative error in MTTR increases relative OEE, while positive error decreases it. Although the magnitude of the effect is smaller than that observed for programmed-time error, the analysis confirms that downtime-recording inaccuracies remain analytically relevant.

E. Data governance improvement

In addition to the statistical reduction in inconsistency, the digital ecosystem changed the governance of the analytical workflow. The comparison below summarizes the operational differences between the 2025 manual process and the 2026 digital process.

TABLE VI. DATA GOVERNANCE COMPARISON TABLE

Dimension	System	
	2025 (manual)	2026 (digital)
Primary capture medium	Paper	Digital Capture Interface
Number of transcriptions per record	2	0
Error traceability	Low or none	High
Error detection time	Late and difficult to find	Immediate and traceable
Type of correction	Manual on paper and Excel	Digital in Microsoft Lists
Consolidation time for analysis	High manual consolidation effort	Automatic update after refresh
OEE calculation	Semi-manual with individual formulas in Excel	Automatic with calculated tables in Power Query
Risk of human error	High	Minimal and easy to correct
Process scalability	Low	High
Repeatability	Requires complete repetition of the process periodically for analysis	Reusable and sustainable
Paper usage	High	Removed

The digital workflow transformed the analysis process from a manually repeated reconstruction effort into a reusable and scalable workflow.

IV. DISCUSSION AND CONCLUSIONS

The results confirm that reliable OEE analysis depends not only on the correctness of the KPI formula, but also on the integrity and structure of the production data pipeline. In the original manual workflow, the data architecture itself

introduced uncertainty, delayed the detection of errors, and distorted the resulting KPI. This issue is especially relevant in flexible manufacturing environments, where even a limited number of inconsistent records can produce significant interpretive errors when key variables are captured incorrectly.

The reduction in source-record inconsistency from 43.43% to 2.25% shows that digital intervention was not incremental, but substantial. The statistical significance of the two-proportions test strengthens this result, while the paired comparison of the 2025 OEE values demonstrates that poor data quality affected not only traceability and administrative effort, but also the numerical value of the indicator itself. In practical terms, decisions based on the original manual dataset would have relied on a distorted analytical representation of system performance.

Reduced source-record inconsistency also provides a mechanism for improving analytical reliability. Reliability in this context refers to the ability of the measurement process to produce consistent and reproducible OEE values from the same operational events. When production, failure, and quality records contain missing, mismatched, or manually reconstructed information, the resulting OEE depends on subjective reconciliation decisions. By reducing inconsistencies and preserving traceability from each operational record to the final indicator, the proposed digital ecosystem decreases the influence of data-processing variability, making the calculated OEE a more dependable representation of the manufacturing process.

The sensitivity analyses provide a mechanistic explanation for this distortion. Programmed time recording error propagates through the performance component by modifying the number of planned pallets, whereas MTTR recording error affects OEE through availability. This distinction is important because it shows that capture errors do not influence OEE uniformly. For the present study, the programmed-time pathway is particularly relevant since it reflects the need to standardize production-planning logic together with data capture and integration.

The study has several limitations. First, the evaluation was conducted in a single educational flexible manufacturing cell, which limits direct generalization to industrial environments. In addition, the digital ecosystem was evaluated against the previous manual workflow rather than an independent reference measurement system, demonstrating improvements in inconsistency, traceability, and repeatability, but not absolute measurement accuracy. Second, the paired OEE comparison was limited to the valid observations for which both original and corrected values were available, while the sensitivity analyses were deterministic and designed to evaluate error propagation rather than stochastic variability.

Finally, the proposed ecosystem was implemented using low-code Microsoft technologies suitable for rapid deployment; although these platforms may present scalability limitations in high-frequency industrial settings, the underlying data architecture and integration workflow are transferable to enterprise-level databases or Manufacturing Execution Systems (MES). The study therefore evaluates

improvements in data integrity, traceability, and analytical reliability, rather than direct increases in physical production performance.

From the perspective of intelligent systems, robotics, and automation, the main contribution of this work is the integration of a physical flexible manufacturing cell with a traceable digital information layer capable of supporting more reliable operational analytics. The paper does not claim that digitalization alone improves production performance. Instead, it demonstrates that digital standardization makes OEE analysis trustworthy enough to reveal deeper process-improvement opportunities that would otherwise remain hidden behind manual transcription, delayed reconciliation, and repeated spreadsheet rework.

Overall, this study presented a digital data-capture and integration ecosystem for standardized, traceable OEE analysis in a flexible manufacturing cell. The results showed that the original manual workflow was characterized by a high inconsistency rate, duplicated effort, weak traceability, and delayed correction. After the implementation of the digital ecosystem, inconsistency was significantly reduced, the OEE values became more analytically dependable, and the workflow changed from a repetitive manual reconstruction process into a reusable and scalable data pipeline.

The main implication of this work is that reliable performance analysis in flexible manufacturing cells depends not only on the calculation formula, but also on standardized data capture, traceable records, and an integrated digital workflow. The proposed ecosystem improved repeatability, traceability, and efficiency of the analysis process; however, the study was limited to one educational cell and did not evaluate direct changes in physical production performance. Future work should validate the ecosystem over additional production cycles, compare automated and manual calculations through an equivalence test, and extend the workflow to other manufacturing cells or processes.

V. AUTHOR'S CONTRIBUTION

I.D.: Study design, data analysis, manuscript drafting.

F.B.M. & K.H.C.: Conceptualization, manuscript review.

P.C.R.V.: Manuscript review, final editing.

J.B.E. & S.A.C.: Project investigation, final review.

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